Cover Letter

21 August 2026

Editors, *Scientific Reports*

Dear Editors,

Please consider “Compartment-Stratified Systematic Reanalysis of Complement, Coagulation, and Matrix Transcriptional Programs in Diabetic Kidney Disease.” The submission systematically screens 322 unique dataset records, prevents double counting of paired compartments and overlapping sources, and separates gene-level random-effects evidence from design-aware study-wise pathway tests. No individual gene met FDR<0.05. Under a common-measurement, studentized maxT analysis, complement, vascular-wall interaction, chemokine-receptor binding and extracellular-matrix organization met an explicitly operational two-of-three-source criterion; coagulation did not. The manuscript makes no causal, protein-activity or universal cross-compartment claim.

This revision directly addresses exchangeability, platform-dependent pathway membership and maxT scaling by using sex-restricted or exact allocations, common measurable genes, studentized maxT, 100,000 Monte Carlo allocations where enumeration was infeasible, bootstrap uncertainty and leave-one-source-out analysis. It adds a complete PRISMA 2020 checklist, source-level risk/confounding and ethics provenance, formal GEO data citations, a comparison with prior DKD integrations, and a self-contained code archive tested from extraction. Unresolved ethics fields are reported transparently rather than inferred.

The work is original and is not under consideration elsewhere. The author declares no competing interests and no specific funding. Only public de-identified data were analyzed. The exact versioned snapshot is https://github.com/denglizhen-113/dkd-focused-universe-reanalysis/tree/v1.2.0; no DOI has been assigned.

Sincerely,

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